TECHNICAL DOCUMENTATION — TIKTOK VIDEO ANALYSIS PIPELINE

Custom Macedonian Video Processing Pipeline

A specialised extension of the main TikTok creator analysis pipeline, designed for targeted processing of Macedonian-language TikTok videos: download, frame extraction, audio extraction, and high-accuracy Whisper transcription.

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1. Overview

The main pipeline processes hundreds of TikTok creators in bulk, downloading up to 3 videos per creator, extracting frames and audio, transcribing with `faster-whisper (tiny)`, and indexing into Elasticsearch and FAISS for semantic search.

The **custom pipeline** is a focused, standalone extension for scenarios where you need to process a specific set of videos with higher quality — in particular, Macedonian-language content that benefits from a larger, language-locked Whisper model. It is fully isolated: all output is stored under `custom_pipeline/` and **nothing in the main pipeline is modified**.

> **Key difference from the main pipeline:** Uses `openai-whisper large` with `language="mk"` (Macedonian) instead of `faster-whisper tiny` with auto language detection — significantly more accurate for Macedonian speech.

2. Pipeline Steps

The pipeline executes four sequential stages. Ray is used for parallel execution of CPU-bound tasks.

1. **URL Parsing**

   Each full TikTok URL (e.g. `https://www.tiktok.com/@dinevv/video/7596645699547106571`) is parsed using a regex to extract the **creator username** and **video ID**. No external API calls or network requests are needed at this stage. Invalid URLs are skipped with a warning.

2. **Download (Ray, parallel)**

   Each video is downloaded as `.mp4` using `yt-dlp` via a `@ray.remote` task. Downloads are retried up to 4 times on failure. If the file already exists it is skipped (checkpointing). Output:

`custom_pipeline/videos/<creator>/<video_id>/<video_id>.mp4`

3

Audio & Frame Extraction (Ray, parallel)

Two parallel Ray tasks run per video after download completes:

- **Audio:** FFmpeg strips the audio track and encodes it as 16kHz mono PCM WAV — the exact format required by Whisper.
- **Frames:** FFmpeg extracts 1 frame per second from seconds 1–10, saving up to 10 PNG images. These can be used for visual analysis or gender classification.

4

Transcription (Ray GPU Actor, Whisper large)

A Ray GPU Actor loads `openai-whisper large` once into VRAM and processes all audio files sequentially. The language is locked to `"mk"` (Macedonian ISO-639-1), eliminating language detection overhead and maximising transcription accuracy. Each transcription is saved as a plain text file: `custom_pipeline/transcriptions/<video_id>.txt`

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3. Output Structure

```
custom_pipeline/
  run_custom.py             ← main script (edit URLS list here)
  documentation.html        ← this document
  videos/
    <creator>/
      <video_id>/
        <video_id>.mp4     ← downloaded video
        <video_id>.wav     ← 16kHz mono audio
        frames/
          <video_id>_frame_01.png
          <video_id>_frame_02.png
          ...               ← up to 10 frames
  transcriptions/
    <video_id>.txt         ← Whisper transcription (Macedonian)
```

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4. Configuration

All configuration lives at the top of `run_custom.py`. The only required edit is the `URLS` list.

Parameter	Default	Description
`URLS`	—	List of full TikTok URLs to process
`WHISPER_MODEL_SIZE`	`"large"`	Whisper model size: tiny / base / small / medium / large / large-v3

Parameter	Default	Description
`DEVICE`	`"cuda"`	Compute device for Whisper — use `"cpu"` if no GPU available
`language`	`"mk"`	ISO-639-1 language code passed to Whisper (Macedonian)

Adding videos

```
URLS = [
    "https://www.tiktok.com/@dinevv/video/7596645699547106571",
    "https://www.tiktok.com/@someuser/video/7234567890123456789",
    # Add as many as needed...
]
```

Upgrading to the strongest model

```
WHISPER_MODEL_SIZE = "large-v3"   # latest, ~3GB, best accuracy for Macedonian
```

5. Technology Stack

Component	Library / Tool	Role
Distributed execution	Ray	Parallel download and extraction tasks; GPU Actor for Whisper
Video download	yt-dlp	Downloads TikTok videos from public URLs
Audio extraction	FFmpeg	Converts video to 16kHz mono WAV
Frame extraction	FFmpeg	Extracts PNG frames at 1fps from first 10 seconds
Transcription	openai-whisper	Large model (1.5B parameters), language-locked to Macedonian

> **Why Ray?** Downloads and extractions are I/O-bound and independent per video — Ray parallelises them automatically across CPU cores. The Whisper Actor pattern ensures the large model is loaded into GPU memory exactly once, regardless of how many videos are processed.

6. Whisper Model Selection & Benchmark Results

The main pipeline uses `faster-whisper tiny` (39M parameters) for bulk throughput across hundreds of creators. During development of this custom module, **tiny** and **small** were both tested on Macedonian audio and found to produce frequent word substitutions and dropped phrases — particularly on conversational speech. This pipeline therefore uses `openai-whisper large` (1.5B parameters), trading speed for substantially higher accuracy.

Word Error Rate (WER) — Published Benchmarks

Sources: OpenWhispr benchmark summary (openwhispr.com), VexaScribe real-world evaluation (novascribe.ai), Whisper-RIR-Mega reverberant speech benchmark (arxiv.org/2603.02252, 2026). WER = (substitutions + deletions + insertions) / total words. Lower is better.

Model	Params	VRAM	English WER (clean)	Multilingual WER	Reverb penalty (Δ WER)	Speed vs large
tiny	39M	~1 GB	~7.6 – 15%	~12%	+15.5 pp	~32×
small	244M	~2 GB	~3.4 – 9%	~7%	+7.4 pp	~6×
medium	769M	~5 GB	~2.9 – 6%	~5%	+5.9 pp	~2×
large ← used here	1.5B	~10 GB	~2.7 – 5%	~4%	+2.3 pp	1×
large-v3	1.5B	~10 GB	~2.4%	~3.5%	+2.3 pp	1×

~12%	**~7%**	**~4%**	**3×**
tiny multilingual WER	small multilingual WER	large multilingual WER	large vs tiny accuracy gain

Why tiny and small were insufficient

The reverberant speech benchmark (Whisper-RIR-Mega, 2026) quantifies how badly each model degrades when audio quality drops — typical for phone-recorded TikTok videos. `tiny` loses **+15.5 percentage points of WER** under reverberation, while `large` loses only **+2.3 pp**. For a lower-resource language like Macedonian — which has less training data than English — the gap widens further. In practice, tiny and small produced intelligible but incomplete transcripts with frequent dropped words on conversational Macedonian audio; large produced near-complete transcripts.

Why medium was not chosen

Medium (769M parameters) achieves ~5% multilingual WER — a meaningful improvement over small, but still approximately **25% more errors than large** on multilingual benchmarks. Given that GPU VRAM is not a bottleneck in this setup and batch sizes are small (targeted per-video processing rather than bulk), the accuracy gain of large over medium justifies the additional compute. Large also shows significantly better robustness to audio noise (+5.9 pp vs +2.3 pp reverb penalty).

> **Conclusion:** For Macedonian TikTok videos — conversational speech, phone microphone quality, lower-resource language — `openai-whisper large` is the minimum recommended model. `large-v3` further reduces multilingual WER by ~12% relative and is the recommended upgrade for maximum accuracy.

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7. Context — Gender Classification Results (Main Pipeline)

The broader pipeline this module extends includes an image-based gender classification component applied across all 933 TikTok creator videos. Results from that component provide context for the quality and reliability of the overall system.

933	91.5%	97.2%	+5.7%
Videos processed	Baseline accuracy (v1)	KD + Late Fusion (v2)	Improvement over baseline

Method	Accuracy	F1 (weighted)	vs Baseline
v1: Hard-label fine-tuned (thresh=0.70)	87.9%	88.0%	−3.6%
v1: Baseline + Majority Vote	91.5%	91.5%	—
v2: KD Fine-tuned + Late Fusion	**97.2%**	**97.2%**	+5.7%

Evaluation on 141 held-out test videos. Gender classification uses MTCNN face detection on the first 4 frames per video and a Vision Transformer (ViT) classifier (`dima806/man_woman_face_image_detection`, 98.7% benchmark accuracy on clean faces). The v2 improvement comes from Knowledge Distillation (soft pseudo-labels, KL divergence loss, temperature T=3) combined with Late Fusion at inference (summing softmax probabilities across frames rather than majority vote).
